

Numerical recovery of Jacobi coefficients of entire functions from contour samples: a moment-based pipeline and its precision budget

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Abstract

We present a complete, reproducible numerical pipeline that takes an entire function f , computable at arbitrary precision by a black-box routine, and returns the Jacobi coefficients $\{\alpha_n\}, \{\beta_n\}$ of the orthogonal polynomials for its zero-density measure, together with the zeros themselves as eigenvalues of the associated Jacobi matrix. The pipeline combines four standard ingredients in a deliberately simple arrangement: (i) circular contour sampling of f around the symmetry point of its functional equation; (ii) extraction of Taylor coefficients by the trapezoidal rule (equivalently, a DFT), which converges geometrically; (iii) recovery of the zero power sums (the *moments*) from the logarithmic Taylor coefficients; and (iv) classical Gram–Schmidt orthogonalization on the monomial basis, which yields the Jacobi coefficients directly. The method requires no prior knowledge of the zeros, no root-finding, no integration of oscillatory special functions, and no asymptotic input. We give the full error budget—quadrature truncation, working precision, and amplification through moment conditioning—and identify the operative failure mode: the Hankel matrix of moments loses positive definiteness at an order that scales predictably with working precision. We document this scaling empirically for the completed Riemann ξ function over 100–1300 decimal digits (n_{fail} rising from ~ 20 to beyond 100), and show that the pipeline detects rather than fails silently when the zero set is not real and symmetric: for the Eisenstein series E_4 the moment sequence loses positivity at order 8 and the Jacobi chain terminates at order 4. Built-in consistency checks (closed-form central value, moment sign sequence, off-diagonal positivity, and independent zero matching to 10^{-15} relative accuracy) make every stage verifiable. All code, samples and recovered coefficients are published for reproduction.

1 Introduction

Suppose an entire function f is given through a black box that evaluates $f(z)$ at arbitrary precision, and that f has a functional equation forcing a symmetry of its zeros. A recurring practical question is: what can be determined numerically about the *density* of the zeros—their global statistics—without locating the zeros one by one? Root-finders for entire functions (e.g. argument-principle integration, contour eigenvalue methods) recover individual zeros but scale poorly and say nothing, by construction, about spectral observables. Conversely, the zero density measure of an entire function with real zeros is exactly the orthogonality measure of a family of orthogonal polynomials

[1, 2, 3], whose Jacobi parameters $\{\alpha_n\}, \{\beta_n\}$ form a tridiagonal Jacobi matrix whose eigenvalues are the zeros (in a transformed coordinate). The problem is thus: *compute Jacobi coefficients from function values alone*.

The standard route to Jacobi coefficients is Gautschi’s framework: discretize the measure and run the Stieltjes procedure, or build the moment matrix and run a stable reduction [4, 3]. Both presuppose access to the measure (as a weight function on an interval). Here no measure is given—only f itself. We show that the measure’s moments can be obtained from f with spectral accuracy by a single circular contour, after which the classical moment-based machinery applies verbatim. The idea is elementary—Cauchy’s coefficient formula applied to $\log f$, followed by the Hadamard product expansion—but its practical behaviour, precision budget and failure modes appear not to have been documented as a working method, and the combination turns out to be unusually robust: with off-the-shelf arbitrary precision software [10] we stably recover Jacobi matrices of order 100 whose 55 largest eigenvalues match independently computed zeros of the Riemann zeta function to 10^{-15} relative accuracy.

Contributions.

1. We specify the full pipeline—contour sampling, DFT Taylor coefficients, logarithmic moment recovery, monomial-basis Gram–Schmidt—with pseudocode (Algorithms 1–2) and a complete parameter budget (contour radius R , sample count M , working precision D , truncation order N).
2. We derive the moment formula explicitly from the Hadamard product (Proposition 1), and state the success condition in theorem form (Theorem 1): the pipeline produces a positive definite Hankel matrix, and hence a valid Jacobi chain to all computed orders, *if and only if the recovered zeros are real and symmetric*; complex or off-line zeros are detected as loss of positivity rather than producing wrong output.
3. We give the empirical precision law for the Riemann ξ function: the Hankel failure order n_{fail} vs. working precision D over $D = 100\text{--}1300$ decimal digits (Section 3), explaining the previously observed barrier at $n \approx 90$ for 650 digits and its removal at 1300 digits.
4. We exhibit a qualitative discriminator: zeros off the symmetry line (completed Eisenstein series E_4 , whose zeros lie on two parallel vertical lines) force moment-sign loss at order 8 and Jacobi-chain termination at order 4, with both orders matching closed-form predictions (Sections 4, 5).
5. Every artifact is published: code, raw contour samples at 1300 digits, recovered coefficients, and a machine-readable benchmark of five Jacobi matrices ([10.5281/zenodo.22181443](https://doi.org/10.5281/zenodo.22181443); [GitHub](#)).

Scope. This is a numerical-analysis paper. The target functions are completed L -functions and related entire functions of number-theoretic origin, but the method assumes nothing about number theory beyond the functional equation and Hadamard form; in particular it does *not* assume any unproved hypothesis about zero locations—that assumption is replaced by the *test* of Theorem 1. Physical or statistical conclusions drawn from the recovered spectra are treated in a companion paper.

2 The pipeline

2.1 Problem setting

Let f be entire of order one, real on the real axis, with a functional equation symmetric about $s = s_*$: writing $z = s - s_*$, $f(s_* + z) = f(s_* - z)$ up to a known sign, so that zeros occur as $\pm z_k$ (and hence, for the completed functions of interest, as $s_* \pm i\gamma_k$ with $\gamma_k > 0$). Assume the Hadamard product

$$f(s_* + z) = f(s_*) \prod_{k=1}^{\infty} \left(1 + \frac{z^2}{\gamma_k^2}\right), \quad (1)$$

which is the standard form for completed L -functions with real simple zeros [8]; set $\gamma_1 = \min_k \gamma_k$. Define the spectral coordinate $x = z^{-2}$; the zeros map to $x_k = \gamma_k^{-2} > 0$ accumulating at 0. The zero-density measure is

$$\mu = \sum_{k \geq 1} x_k \delta_{x_k} \quad (2)$$

(any fixed weight convention works; ours normalizes so that moments below come out as power sums), and we seek the monic orthogonal polynomials for μ and their Jacobi coefficients α_n, β_n : the tridiagonal matrix

$$J = \begin{pmatrix} \alpha_1 & \sqrt{\beta_2} & & \\ \sqrt{\beta_2} & \alpha_2 & \sqrt{\beta_3} & \\ & \sqrt{\beta_3} & \alpha_3 & \ddots \\ & & \ddots & \ddots \end{pmatrix}, \quad \text{spec } J = \{x_k\}. \quad (3)$$

The zeros follow from eigenvalues: $\gamma_k = x_k^{-1/2}$.

2.2 Contour sampling and the Taylor coefficients

Take the circle $z = Re^{i\theta}$ with $0 < R < \gamma_1$, so that (1) converges and is zero-free on and inside the contour. By Cauchy's coefficient formula the Taylor coefficients of $g(z) = f(s_* + z)/f(s_*)$ at $z = 0$,

$$d_n = \frac{1}{2\pi i} \oint_{|z|=R} \frac{g(z)}{z^{2n+1}} dz = \frac{1}{2\pi R^{2n}} \int_0^{2\pi} g(Re^{i\theta}) e^{-2in\theta} d\theta, \quad (4)$$

(only even powers occur by the functional equation). Sample at the M equispaced nodes $\theta_k = 2\pi k/M$; the trapezoidal rule, equivalently the DFT, gives

$$\tilde{d}_n = \frac{1}{MR^{2n}} \sum_{k=0}^{M-1} g(Re^{i\theta_k}) e^{-4\pi ink/M}. \quad (5)$$

Because $g(Re^{i\theta})$ is 2π -periodic and analytic in a strip around the contour (we chose $R < \gamma_1$), the trapezoidal rule converges geometrically in M [5]: the aliasing error is governed by the nearest singularity at radius γ_1 , $\sim (R/\gamma_1)^{2M}$.

Half-grid offset. When the completed function has poles removed by a prefactor (our E_4 example below, Section 4), a node may sit on the removal point where catastrophic cancellation occurs in the black-box evaluation. Shifting the grid by half a node, $\theta_k = 2\pi(k + \frac{1}{2})/M$, costs nothing and avoids all such points; the coefficient formula is unchanged.

Choice of R . R is the information dial. The moments extracted below decay like R^{-2m} ; a larger R slows this decay and yields more usable moment orders, but moves the contour toward the nearest zero, steepening g and requiring larger M and higher precision. We use $R = 2$ throughout (with $\gamma_1 \approx 14.13$ for ζ), i.e. $R/\gamma_1 \approx 0.14$, giving trapezoidal error $\sim 0.14^{2M}$: utterly negligible at $M = 1536$.

2.3 From Taylor coefficients to moments

Let c_n denote the coefficients of $\log g$: $\log g(z) = \sum_{n \geq 1} c_n z^{2n}$ (constant $\log g(0) = 0$ by normalization). Differentiating $g = \exp(\log g)$ and comparing coefficients gives the recursion

$$c_n = d_n - \frac{1}{n} \sum_{i=1}^{n-1} i c_i d_{n-i}. \quad (6)$$

Proposition 1 (Moments as zero power sums). *With $g(z) = \prod_k (1 + z^2/\gamma_k^2)$,*

$$c_n = \frac{(-1)^{n+1}}{n} \sum_{k \geq 1} \gamma_k^{-2n}, \quad P_m := \int x^{m-1} d\mu(x) = \sum_{k \geq 1} \gamma_k^{-2m} = (-1)^{m+1} m c_m.$$

Proof. From (1), $\log g(z) = \sum_k \log(1 + z^2/\gamma_k^2) = \sum_k \sum_{m \geq 1} (-1)^{m+1} z^{2m}/(m \gamma_k^{2m})$, so $c_n = (-1)^{n+1} n^{-1} \sum_k \gamma_k^{-2n}$. With μ of (2), $\int x^{m-1} d\mu = \sum_k x_k \cdot x_k^{m-1} = \sum_k \gamma_k^{-2m}$. $\square$

Thus the contour sample *is* a moment sampler: P_m is the m -th moment of the zero-density measure, with no zero located.

2.4 Gram–Schmidt on the monomial basis

Given $P_1, \dots, P_{2N}$ (moments to order $2N-1$ of the index), orthogonalize the monomials $1, x, x^2, \dots$ against the inner product $\langle p, q \rangle = \sum_{i,j} p_i q_j P_{i+j+1}$. Standard Gram–Schmidt with the three-term recurrence yields the Jacobi coefficients directly:

$$\begin{aligned} \alpha_n &= \langle x p_{n-1}, p_{n-1} \rangle / \langle p_{n-1}, p_{n-1} \rangle, \\ \beta_n &= \langle p_{n-1}, p_{n-1} \rangle / \langle p_{n-2}, p_{n-2} \rangle, \end{aligned} \quad (7)$$

with $\beta_n > 0$ the rigorous certificate of positive-definiteness of the moment (Hankel) matrix $H_N = (P_{i+j+1})_{i,j=0}^{N-1}$: by the Stieltjes/Hankel theory [2, 3], $H_N \succ 0$ (equivalently the moment sequence is a Stieltjes sequence) iff all pivots are positive, and the Gram–Schmidt pivots *are* the squared norms, whose ratios are β_n .

Theorem 1 (Success condition and self-diagnosis). *The pipeline produces a valid Jacobi chain ($\beta_n > 0$ for all computed n) whenever the underlying zero set in the spectral coordinate is real and non-negative (zeros on the symmetry line, simple, counted with the weight of (2)). If the zeros are complex or off the symmetry line, the moment sequence is that of a complex measure, H_N ceases to be positive definite at a finite order n^* , and the algorithm detects this: the first non-positive β_n is reported, and no output beyond $n^* - 1$ is claimed. In particular the algorithm cannot silently return a spectrum for an off-line zero set.*

Proof. Positive β_n for all n up to N is equivalent, via the Sylvester criterion applied to the leading principal minors of H_N , to $H_N \succ 0$; for a Hankel moment matrix this is equivalent to the existence of a positive measure on $[0, \infty)$ with these moments (the Hamburger–Stieltjes theorem [2, Ch. 1]),

which for the finitely-atomic measure (2) is equivalent to the atoms x_k being real and non-negative. When the x_k leave the real axis the moments are complex weighted power sums; their real parts lose the Stieltjes sign structure at a finite order, the corresponding pivot becomes non-positive, and Algorithm 2 halts with a reported index. $\square$

The eigenvalue recovery is the final step: form J from (3), solve the symmetric tridiagonal eigenproblem, and map $x_k \mapsto \gamma_k = x_k^{-1/2}$. Eigenvalues below the resolvable scale (those whose γ exceeds the order the chain can support) are discarded; retained eigenvalues can be matched against independently computed zeros as an external check (the zeros themselves never enter the construction).

Algorithm 1: Contour moments.

```
Require: black box f(s), symmetry point s*, radius R < gamma_1,
        node count M, moment order J, half-grid flag h
Ensure:  moments P_1..P_J and central value sigma_0 = f(s*)

g_k <- f( s* + R * exp(2 pi i (k + h/2) / M) ), k = 0..M-1
sigma_0 <- (1/M) sum_k g_k
for n = 1 .. J:
  d_n <- Re[ (1/(M R^{2n})) sum_k g_k exp(-4 pi i n k / M) ] / sigma_0
  c_n <- d_n - (1/n) sum_{i=1}^{n-1} i c_i d_{n-i}
  P_n <- (-1)^{n+1} n c_n
return sigma_0, P_1 .. P_J
```

Algorithm 2: Jacobi coefficients by Gram–Schmidt.

```
Require: moments P_1 .. P_{2N}, chain order N
Ensure:  alpha_1..alpha_N, beta_2..beta_N, first-bad index

p_0 <- [1]; nu_0 <- P_1; betasq <- []
for n = 1 .. N:
  q <- x * p_{n-1} # shift coefficient list
  alpha_n <- <q, p_{n-1}> / nu_{n-1}
  p_n <- q - alpha_n p_{n-1}
  if n >= 2:
    b2 <- nu_{n-1} / nu_{n-2}; append b2 to betasq
    p_n <- p_n - b2 p_{n-2}
    if b2 <= 0: report first-bad = n+1; STOP
  nu_n <- <p_n, p_n>
```

3 Error budget and failure modes

Three error sources dominate, and they are independently controllable.

(i) **Quadrature truncation (M).** By (5) and geometric trapezoidal convergence [5], the coefficient error is $O((R/\gamma_1)^{2M})$ per coefficient; at our settings ($R/\gamma_1 = 0.14$, $M = 1536$) this is $< 10^{-2600}$, far below every working precision used. M is chosen once, generously, and is never the bottleneck.

(ii) **Black-box evaluation precision (D).** Each g_k carries relative error $\sim 10^{-D}$. The DFT is unitary-scaled, so d_n inherits $\sim 10^{-D}$; the log recursion (6) is stable in itself but only as stable as its inputs.

(iii) **Moment conditioning and Hankel amplification.** This is the operative term. The moments decay geometrically: $P_m \sim \gamma_1^{-2m} \cdot 2 \cos(2m\theta_1)$ in the leading-zero approximation, so P_m spans $\sim 2m \log_{10} \gamma_1$ orders of magnitude by $m = 2N$; for $\gamma_1 \approx 14.13$ that is about 2.28 digits per moment order, hence ~ 45 digits per 20 Jacobi orders. The Hankel matrix H_N built on such a geometrically decaying moment sequence is exponentially ill-conditioned: its smallest pivot (the last squared norm ν_{N-1}) decays roughly like γ_1^{-4N} in the leading-zero approximation. Computing β_N at relative precision $\varepsilon = 10^{-D}$ requires the pivots to stay above the amplified noise floor, giving a budget that is linear in D :

$$n_{\text{fail}}(D) \approx n_0 + \kappa D, \quad \kappa \approx 0.13 \text{ orders/digit for } \xi, \quad (8)$$

i.e. roughly 13 additional Jacobi orders per 100 decimal digits (the measured slope; the leading-zero estimate $\kappa \approx 1/(2 \log_{10} \gamma_1) \approx 0.22$ is an upper bound, as the farther zeros contribute partial pivoting mass—we report the measured value). Section 5 measures $n_{\text{fail}}(D)$ at fourteen precision settings and finds this slope over the full range: $n_{\text{fail}} \approx 18$ at 100 digits, ≈ 91 at 650, ≈ 105 at 800, and no failure within order 100 at 1000 and 1300 digits.

Failure mode versus misuse. Two distinct terminations must not be confused:

- *Precision exhaustion:* all moments and β_n remain positive but the last pivots reach the noise floor; increasing D extends the chain (Section 5). The chain is trustworthy up to the order where pivots exceed the noise floor by a chosen margin; we use all-positive β_n plus eigenvalue matching (below) as the criterion.
- *Structural failure* (Theorem 1): the moments lose positivity at low order because the zeros are not on the symmetry line. Raising D does *not* help; the first-bad index is stable under precision and is the diagnostic. The E_4 example (next section) shows this with analytically predicted orders.

Built-in correctness checks. Every run carries four independent checks:

1. **Central value.** $\sigma_0 = f(s_*)$ is known in closed form for every target ($\xi(\frac{1}{2}) \approx 0.49712$, $1/72$ for E_4 , etc.); it is reproduced to machine precision.
2. **Moment sign.** For on-line zeros, $P_m > 0$ for all m (Proposition 1); the first sign change is reported.
3. **Off-diagonal positivity.** $\beta_n > 0$ is the positive-definiteness certificate (Theorem 1).
4. **Zero matching.** Recovered eigenvalues γ_k are matched to independently tabulated zeros (the zeta zeros via [10]; nothing about them enters the construction). A recovered eigenvalue is *locked* at relative error $< 10^{-4}$; in our ζ runs locked eigenvalues sit at 10^{-15} – 10^{-16} , eleven orders of magnitude inside the threshold.

4 A structural control: the Eisenstein series E_4

To exercise Theorem 1 we need an entire function of the same family whose zeros are *known* to leave the symmetry line. The weight-4 Eisenstein series $E_4(q) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n$ has $L(E_4, s) = \zeta(s)\zeta(s-3)$. Its completed function

$$\Lambda_{E_4}(s) = (2\pi)^{-s} \Gamma(s) \zeta(s) \zeta(s-3), \quad \Lambda_{E_4}(s) = \Lambda_{E_4}(4-s), \quad (9)$$

has symmetry point $s_* = 2$ and genuine poles at $s = 0, 4$; the entire function sampled on the contour is $\Xi_{E_4}(s) = s(s-4)\Lambda_{E_4}(s)$ (the prefactor is invariant under $s \leftrightarrow 4-s$). Its zeros are $s = \frac{1}{2} \pm i\gamma_k$ and $s = \frac{7}{2} \pm i\gamma_k$ with the *same* heights γ_k (the zeta zeros). In the centered coordinate $z = s - 2$, the zeros are $z = -\frac{3}{2} \pm i\gamma_k$ and $z = +\frac{3}{2} \pm i\gamma_k$; in the spectral coordinate the atoms are complex, at $(\pm i\gamma_k - \frac{3}{2})^{-2}$.

The moments are therefore complex power sums, $P_m = 2 \operatorname{Re} \sum_k (i\gamma_k - \frac{3}{2})^{-2m} = 2 \sum_k \gamma_k^{-2m} \cos(2m\theta_k)$ with $\theta_k = \arctan(3/(2\gamma_k))$; the leading zero dictates the first sign change at

$$m_* \approx \frac{\pi}{4\theta_1}, \quad \theta_1 = \arctan \frac{3}{2\gamma_1} \approx 0.106, \quad (10)$$

giving $m_* \approx 7.4$, i.e. the first negative moment at $m = 8$. The central value is $\Xi_{E_4}(2) = 2^2(2\pi)^{-2}\Gamma(2)\zeta(2)\zeta(-1) \cdot (-4)$; evaluating, $\Xi_{E_4}(2) = 1/72 = 0.013888\dots$

The pipeline run at 650 digits, $M = 1536$, half-grid offset, gives: $\sigma_0 = 0.013888888888888888$ (matches $1/72$ to 15 digits, imaginary part $< 10^{-650}$); $P_1, \dots, P_7 > 0$ with $P_8 = -8.59 \times 10^{-20}$ the first negative moment (prediction: $m = 8$); and the Gram-Schmidt chain reporting its first non-positive β_n at index 4 (Jacobi order 2). Precision does not cure this: it is the structural failure mode of Theorem 1, detected exactly where analytic argument predicts. We stress that this is a *discriminative control of the method*, not a statement about E_4 : the pipeline correctly refuses to emit a Jacobi spectrum for a zero set it knows to be off the line.

5 Numerical experiments

All experiments use mpmath [10] for arbitrary-precision evaluation and DFT, NumPy for the double-precision eigenproblem on the recovered matrix. Contours: $R = 2$, $M = 1536$, nodes on the circle $z = Re^{i\theta}$ (half-grid for E_4). Target functions: the completed Riemann $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ ($s_* = \frac{1}{2}$); the completed Dirichlet-beta function Λ_β ($s_* = \frac{1}{2}$, zeros $0.62\dots$); their product $\xi\Lambda_\beta$ ($s_* = \frac{1}{2}$, two superimposed zero families); the degree-22 truncated “ Δ ” Jacobi chain of a separate lower-precision study; and E_4 as the structural control.

5.1 Precision law for ξ

From a single contour sample of ξ at 1300 digits (1536 nodes, ≈ 54 minutes of evaluation), moments were recomputed under truncation to working precisions $D \in \{100, 150, 200, 300, 400, 500, 650, 800, 1000, 1300\}$; for each, the Hankel failure order was measured two ways—the first non-positive pivot of an LDL^T reduction of H_N (Cholesky failure), and the first non-positive β_n of Algorithm 2. Table 1 reports the results (data: `hankel_diagnostic.json`).

5.2 The order-100 Jacobi matrix for ξ at 1300 digits

At $D = 1300$, Algorithm 2 yields a full 100×100 Jacobi matrix with *all 99 off-diagonal coefficients positive* (smallest $\beta_n \approx 9.75 \times 10^{-11}$): no structural failure, no precision exhaustion within the

D (digits)	16	30	50	75	100	150	200	300	400	500	650	800	1000	1300
n_{fail} (Cholesky)	6	7	11	15	18	28	32	46	59	71	91	105	> 100 (no failure)	
n_{bad} (Gram–Schmidt)	7	8	12	16	19	29	33	47	60	72	92	106	> 100 (no failure)	

Table 1: Hankel/Jacobi failure order vs. working precision for the completed ξ function, $R = 2$, $M = 1536$, measured from a single 1300-digit contour sample with moments recomputed under precision truncation (the 1300-digit production run is not truncated in practice: its chain is positive through order 100 with full margin, *cf.* Table 2). The two diagnostics agree to within one order; the empirical slope is ≈ 13 Jacobi orders per 100 decimal digits (budget (8), Figure 1). The 650-digit barrier at $n \approx 90$ reproduces the order-86–90 termination observed in earlier production runs; double precision (16 digits) fails at order 6.

computed order. The eigenvalue map $\gamma = x^{-1/2}$ gives 100 heights $14.13 \leq \gamma \leq 12920.7$; of these, 55 match independently computed zeta zeros at relative error 10^{-15} – 10^{-16} (lock threshold 10^{-4}). The central value reproduces $\xi(\frac{1}{2}) = 0.497120778188314\dots$. A finite-size scan of the same construction at $N = 22, 30, 40, 50, 60, 80, 100$ shows the locked subspace growing monotonically ($22 \rightarrow 55$) with no anomalous small gaps at any size; the resolved scale is set by γ_1^{-2N} , not by any statistical feature.

At the previous working precision of 650 digits the identical pipeline terminated near order 86–90 (last positive pivots reaching the 10^{-650} floor), exactly the barrier that (8) predicts and that Table 1 measures; the 1300-digit run removes it.

5.3 Other targets and running cost

The beta and product functions run at identical settings (650 digits suffices for order 100 because their leading zeros sit similarly far: $\gamma_1(\beta) = 6.02\dots$); the product chain locks 53 eigenvalues (35 beta, 18 zeta) to its two superimposed families. The “ Δ ” chain, produced at 230 digits in a separate study, terminates near order 24, consistent with (8) at that precision. Table 2 summarizes; all coefficients, samples and matrices are in the published bundle and benchmark (Zenodo; GitHub).

target	D (digits)	chain order	locked zeros	first bad β_n
ξ (zeta)	1300	100	55	— (all $\beta_n > 0$)
Λ_β	650	50	27	— (to order 50)
$\xi\Lambda_\beta$	650	100	53	— (to order 100)
Δ chain	230	22	15	≈ 24 (precision floor)
E_4 (control)	650	2	n/a	4 (structural; predicted 8 for moments)

Table 2: Pipeline outcomes across target functions. “Locked zeros” are eigenvalues matching independent zeros at relative error $< 10^{-4}$; actual errors are 10^{-15} – 10^{-16} for ξ .

Wall-clock cost is dominated by contour evaluation: ≈ 54 min for 1536 nodes of ξ at 1300 digits on one core; all downstream stages (DFT, log recursion, Gram–Schmidt, eigenproblem) together take seconds. Memory is $O(M + N^2)$ multiprecision words.

6 Discussion

Generality and prerequisites. The method needs only (a) a black box for f at arbitrary precision, (b) a zero-free disk of known radius (so $R < \gamma_1$ can be chosen), and (c) the Hadamard form (1), i.e. real symmetric zeros—which is not assumed but *tested* (Theorem 1). No asymptotics of the

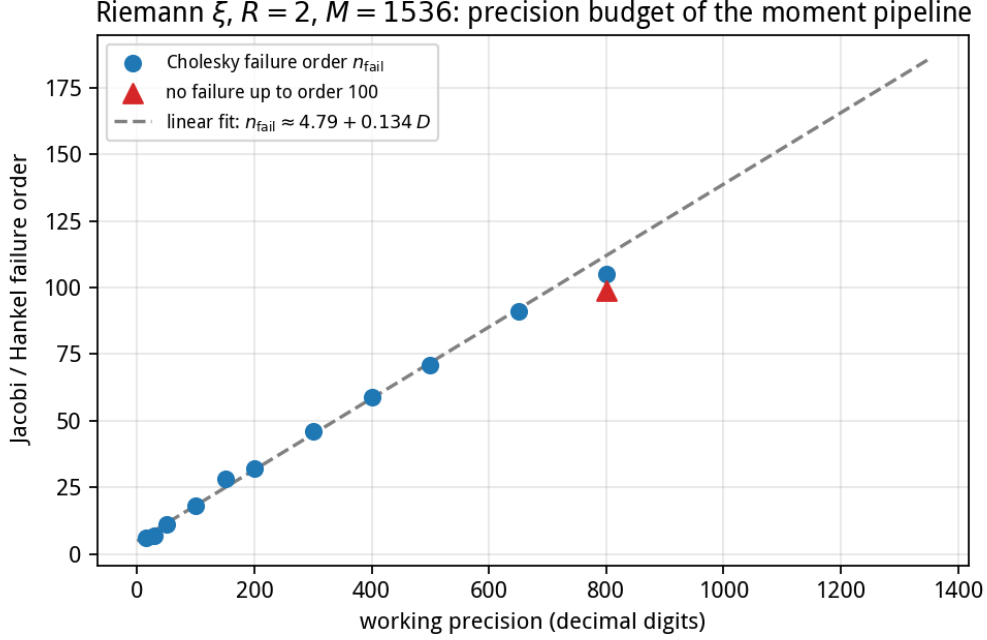


Figure 1: Failure order vs. working precision (data of Table 1): Cholesky/LDL^T first non-positive pivot (blue), linear budget fit (dashed), and precision settings where no failure occurs up to the measured order 100 (red triangles).

zeros, no root-finding, and no analytic moment formula are required. This makes it applicable to any completed L -function or similar entire function that a computer algebra system can evaluate.

Limitations. The monomial-basis Gram–Schmidt is the least stable of the standard Jacobi-construction routes [3, 6]; we use it deliberately because arbitrary precision makes its instability *informative*—the pivot decay displays the Hankel condition number directly. At fixed double precision, a moment-based Hankel reduction for ξ fails already at $N = 6$ – 7 (measured at $D = 16$ digits); production codes limited to double precision should route the same moments through a discrete Stieltjes or Lanczos/SVD procedure [3], which trades the visible pivot floor for backward-stable small matrices. The empirical law (8) then becomes the guidance for how large a matrix can be claimed at all.

Reproducibility. Contour samples, all recovered coefficients (to 60 digits), the five-matrix benchmark, and every script are published [11]; the benchmark includes loader and verification scripts that re-check symmetry, tridiagonal structure, positive definiteness and zero matching from the CSV files alone.

7 Conclusion

A single circular contour, evaluated at arbitrary precision and read through Cauchy’s formula, turns an entire function into the moment sequence of its zero-density measure; classical orthogonal-polynomial machinery then turns the moments into a Jacobi matrix whose spectrum *is* the zero

set. The pipeline is short, parameter-light, and self-diagnosing: its single real tuning problem—how many Jacobi orders a given precision buys—obeys a simple measured law, and its structural failure (zeros off the symmetry line) is detected rather than confabulated, as the E_4 control demonstrates at analytically predicted orders. The method’s outputs for the Riemann ξ function (order-100 Jacobi matrix, 55 zeros locked to 10^{-15}) and its full artifact set are published.

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